DEFECT DETECTION IN MOVING FIBER-CONTAINING STRUCTURES

TECHNICAL FIELD

[0001] This application relates to materials technology in general and more specifically to the preparation, processing and detection of fiber-containing structures such as threadlines, braidlines and wirelay structures. More particularly, this application discloses methods and apparatuses for real-time detection and characterization of defects in fiber-containing structures that are in linear motion.

BACKGROUND OF THE INVENTION

[0002] When producing or processing fiber-containing structures, such as threadlines (e.g., multi-filament threads), braidlines (e.g., braided yarns, braided core-sheath structures, etc.) and wirelay structures (e.g., wire lay ropes), defects can occur due to, for example, the presence of broken filaments and impurities.

[0003] Broken filaments, for example, can occur due to tensioning, bending or twisting of a fiber-containing structure. When broken filaments are present on the surface of a fiber-containing structure undergoing in-line processing, such broken filaments can weaken the fiber-containing structure and can also become detached and then reattach at different locations along the fiber-containing structure leading to additional defects. Broken fibers, whether attached or detached from their original filaments, can change in shape and grow in size during subsequent in-line processing.

[0004] Although the terminology of defects relating to fiber-containing structures can vary widely in the relevant art, the defects resulting from broken filaments and impurities generally fall within two categories. The first category relates to broken filaments (either single filaments or groups of filaments) that extend outward from the surface of the fiber-containing structure as branch-like structures—which are often termed as “fluff” or “peel” defects. The second category relates to broken (detached) fibers, or other impurities, that become attached to the surface of the fiber-containing structure and tend to grow into mound-like structures—which are often termed as “pill” or “slub” defects—during in-line processing.

[0005] Because the defects described above can adversely affect the strength and utility of fiber-containing structures, commercially-available products are generally inspected and often rated based on the quantity or concentration of defects. Although the process of rating fiber-containing structures is often performed by human inspectors using magnification devices such as microscopes, manual inspection can be a tedious process that is not well suited to large-scale production. Furthermore, due to the limitations associated with human inspection, real-time corrective intervention to reduce or eliminate the formation of defects is rarely carried out using human inspection during high-speed production or processing of fiber-containing structures.
Documents DE29521575U1 and US3461299A both relate to opto-electronic devices for detecting defects in yarns or similar fibre-containing elongated structures.

SUMMARY OF THE DISCLOSURE

[0006] The present inventors have recognized that a need exists to discover methods for reliably detecting defects in fiber-containing structures while in linear motion during production or processing. A need also exists for such methods and apparatuses to enable the differentiation and characterization of the defects, and to either modify or terminate the production or processing of fiber-containing structures in order to reduce the occurrence of defects. According to the invention, the method defined in claims 1 and 2 provides a solution to the shortcomings of the prior art.

[0007] The following disclosure describes methods and apparatuses for real-time detection of defects in fiber-containing structures that are in linear motion, as well as fiber-containing structures obtained using these methods and apparatuses.

[0008] Embodiments of the present disclosure, described herein such that one of ordinary skill in this art can make and use them, include the following:

(1) One aspect , beyond the scope of the claimed invention, relates to methods for detecting defects in fiber-containing structures by linearly passing a fiber-containing structure through at least one defect detector, measuring at least one cross-sectional diameter of the fiber-containing structure with the defect detector to obtain at least one diameter signal of diameter versus length of the fiber-containing structure, optionally signal processing the diameter signal to obtain at least one signal-processed diameter signal, and comparing the diameter signal, the signal-processed diameter signal, or combinations thereof, against at least one reference signal to generate at least one surface defect signal of a surface defect output versus the

length of the fiber-containing structure, wherein (a) the defect detector measures the cross-sectional diameter of the fiber-containing structure by transmitting light onto the fiber-containing structure with a projector, detecting a silhouette image of the fiber-containing structure with an optical receiver, and calculating the cross-sectional diameter based on a reduction in an amount of the light detected by the optical receiver relative to a total amount of the light transmitted by the projector, and (b) at least one of the defect detector is situated in series with an extrusion apparatus configured to form the fiber-containing structure, a braiding machine configured to form the fiber-containing structure, a tensioning assembly configured to apply tension to the fiber-containing structure, a finish applicator configured to apply a coating to the fiber-containing structure, a godet roll assembly configured to stretch the fiber-containing structure, a winding assembly configured to wind the fiber-containing structure onto a bobbin, or combinations thereof;

(2) Another aspect relates to defect-detected fiber-containing structures obtained by performing the methods (1) described above; and

(3) Another aspect relates to apparatuses for detecting defects in fiber-containing structures, the apparatuses comprising (A) an extrusion apparatus configured to form a fiber-containing structure, a braiding machine configured to form the fiber-containing structure, a tensioning assembly configured to apply tension to the fiber-containing structure, a finish applicator configured to apply a coating to the fiber-containing structure, a godet roll assembly configured to stretch the fiber-containing structure, a winding assembly configured to wind the fiber-containing structure onto a bobbin, or combinations thereof, (B) a defect detector configured to measure at least one cross-sectional diameter of the fiber-containing structure by transmitting light onto the fiber-containing structure with a projector, detecting a silhouette image of the fiber-containing structure with an optical receiver, and calculating the cross-sectional diameter based on a reduction in an amount of the light detected by the optical receiver relative to a total amount of the light transmitted by the projector; and (C) a processor configured to compare at least one diameter signal obtained from the defect detector, optionally at least one signal-processed diameter signal, or combinations thereof, against at least one reference signal to obtain at least one surface defect signal of a surface defect output

versus length of the fiber-containing structure, wherein the defect detector measures the at least one cross-sectional diameter while the fiber-containing structure is linearly passing through the defect detector.

[0009] Additional objects, advantages and other features of the present disclosure will be set forth in part in the description that follows and in part will become apparent to those having ordinary skill in the art upon examination of the following or may be learned from the practice of the present disclosure. The present disclosure encompasses other and different embodiments from those specifically described below, and the details herein are capable of modifications in various respects without departing from the present disclosure. In this regard, the description herein is to be understood as illustrative in nature, and not as restrictive.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Details of this disclosure are explained in the following description in view of figures that show:

[0011] FIG. 1A illustrates a non-twisted, non-braided threadline formed of a plurality of filaments or filament-containing strands;

[0012] FIG. 1B illustrates a braided fiber-containing structure that is formed of a plurality of filaments or filament-containing strands that are braided together;

[0013] FIG. 1C illustrates a core-sheath fiber-containing structure containing a core of filaments or filament-containing strands surrounded by a sheath formed of a plurality of filaments or filament-containing strands that are braided together;

[0014] FIG. 1D illustrates a wirelay fiber-containing structure containing a core of filaments or filament-containing strands surrounded by a wire-laid cover formed of a plurality of filaments or filament-containing strands that are wire-laid together in the same direction;

[0015] FIG. 2A illustrates a fiber-containing structure with a defect in the form of a “fluff” or “peel” caused by a broken filament;

[0016] FIG. 2B illustrates a fiber-containing structure with a defect in the form of a small "pill" or "slub" caused by an impurity attached to the surface of the fiber-containing structure;

[0017] FIG. 2C illustrates a fiber-containing structure with a defect in the form of an elongated "pill" or "slub" caused by built-up of impurities attached to the surface of the fiber-containing structure;

[0018] FIG. 3 illustrates a defect detector configured to detect and measure the cross-sectional diameter of a fiber-containing structure;

[0019] FIG. 4A illustrates a light projecting slit of a defect detector with belt-shaped parallel light passing through the light projecting slit;

[0020] FIG. 4B illustrates a light receiving slit of a defect detector with partially-blocked parallel light passing through the light receiver slit;

[0021] FIG. 5A illustrates a light receiving slit of a defect detector with parallel light partially blocked by a defect-free portion of a fiber-containing structure that is linearly passing through the defect detector;

[0022] FIG. 5B illustrates a light receiving slit of a defect detector with parallel light partially blocked by a defect-containing portion of a fiber-containing structure that is linearly passing through the defect detector;

[0023] FIG. 6A illustrates a moving fiber-containing structure with a peel defect caused by a broken filament, and shows the location of cross-sectional measurements of the fiber-containing structure occurring at regular intervals;

[0024] FIG. 6B illustrates a moving fiber-containing structure with a small slub defect caused by an attached impurity, and shows the location of cross-sectional measurements of the fiber-containing structure occurring at regular intervals;

[0025] FIG. 6C illustrates a moving fiber-containing structure with an elongated slub defect caused by built-up impurities, and shows the location of cross-sectional

measurements of the fiber-containing structure occurring at regular intervals;

[0026] FIG. 7 illustrates an apparatus for producing or processing a fiber-containing structure;

[0027] FIG. 8 illustrates an apparatus for braiding and detecting a fiber-containing structure; and

[0028] FIG. 9 illustrates a defect detector array including two biaxial defect detectors arranged in series relative to a fiber-containing structure moving linearly through the defect detector array.

DETAILED DESCRIPTION

[0029] Details of this disclosure include various methods for detecting defects in moving fiber-containing structures, apparatuses for carrying out the defect-detection methods, and defect-detected fiber-containing structures obtained by performing the defect-detection methods described herein. Certain non-limiting applications for the defect-detection methods of the present disclosure are also described herein.

[0030] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by persons of ordinary skill in the relevant art. In case of conflict, the present specification, including definitions, will control.

[0033] The use of “a” or “an” to describe the various elements and components herein is merely for convenience and to give a general sense of the disclosure. This description should be read to include one or at least one and the singular also includes the plural

unless it is clear that it is otherwise intended.

[0034] Unless expressly stated to the contrary, "or" and "and/or" refers to an inclusive and not to an exclusive. For example, a condition A or B, or A and/or B, is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

[0035] The terms "about" and "approximately" as used herein refer to being nearly the same as a referenced amount or value, and should be understood to encompass ± 5% of the specified amount or value.

[0036] The term "substantially" as used herein, unless otherwise defined, means all or almost all or the vast majority, as would be understood by the person of ordinary skill in the context used. It is intended to take into account some reasonable variance from 100% that would ordinarily occur in industrial-scale or commercial-scale situations.

[0037] Throughout the present description, unless otherwise defined and described, technical terms and methods employed to determine associated measurement values are in accordance with the description of ASTM D855 / D885M – 10A (2014), Standard Test Methods for Tire Cords, Tire Cord Fabrics, and Industrial Filament Yarns Made From Man-made Organic-base Fibers, published October2014.

[0039] The materials, methods, and examples herein are illustrative only and, except as specifically stated, are not intended to be limiting. Methods and materials similar or equivalent to those described herein may also be used in the practice of the present disclosure.

Methods for Detecting Defects in Moving Fiber-Containing Structures

[0040] Disclosed herein are methods for detecting defects in fiber-containing structures that are in linear motion. The term "fiber-containing structure" includes any cord-like structure formed of fibers and/or filaments, such as threadlines, braidlines, wirelays and core-sheath structures.

[0041] **FIGs. 1A** thru **1D** illustrate examples of fiber-containing structures that may be used in methods of the present disclosure.

[0042] **FIG. 1A** illustrates an example of a threadline cord **5** comprising a plurality of filaments (or filament-containing strands) **10** arranged as a non-twisted, non-braided bundle. **FIG. 1B** illustrates an example of a braidline cord **15** comprising a plurality of filaments (or filament-containing strands) **20** arranged as a braided bundle with no central core. **FIG. 1C** illustrates an example of a braided cord **25** having a core-sheath structure comprising a braided sheath **30**, formed of a plurality of braided filaments (or braided filament-containing strands) **32**, surrounding a core **35** formed of filaments (or filament-containing strands). **FIG. 1D** illustrates a wirelay fiber-containing structure **40** having a core-sheath structure comprising a wirelay sheath **45**, formed of a plurality of braided filaments (or braided filament-containing strands) **50**, surrounding a core **55** formed of filaments (or filament-containing strands).

[0043] As explained above, certain defects can occur when producing or processing fiber-containing structures due to, for example, damaged filaments or impurities that can become attached to the fiber-containing structures.

[0044] **FIG. 2A** illustrates a generic fiber-containing structure **60** with a defect **65** in the form of a "fluff" or "peel" caused by a broken filament. Broken filaments can occur, for example, due to tensioning, bending or twisting of a fiber-containing structure. Typically, broken filaments extend outward at an acute angle **70** relative to the direction of travel **75** of the fiber-containing structure **60** in linear motion during processing. However, in some instances the broken filaments may extend outward at an obtuse angle relative to the direction of travel of the fiber-containing structure. In some situations, the broken filaments may include two branch-like structures including a leading-edge branch **67** extending outward at an obtuse angle (shown in **FIGs. 5A** and **5B**) and a trailing-edge branch **65** extending outward at an acute angle (shown in **FIG. 2A**) relative to the direction

of travel.

[0045] **FIG. 2B** illustrates a generic fiber-containing structure **80** with a defect **85** in the form of a "pill" or "slub" caused by an impurity attached to the surface of the fiber-containing structure **80**. The example of **FIG. 2B** illustrates a small (recently-introduced) impurity in which the point of attachment **87** of the "pill" or "slub" occupies a relatively small area compared to the overall size of the "pill" or "slub".

[0046] **FIG. 2C** illustrates a generic fiber-containing structure **90** with a defect **95** in the form of an elongated or enlarged "pill" or "slub" caused by the built-up or shaping of impurities attached to the surface of the fiber-containing structure **90**. The example of **FIG. 2C** illustrates an elongated or enlarged impurity in which the point of attachment **97** of the "pill" or "slub" occupies a relatively large area compared to the overall size of the "pill" or "slub".

[0047] Defect detection methods of the present disclosure employ defect detectors configured to measure at least one cross-sectional diameter of moving fiber-containing structures.

[0048] Such methods may include the steps of (1) linearly passing a fiber-containing structure through at least one defect detector; (2) measuring at least one cross-sectional diameter of the fiber-containing structure with the defect detector to obtain at least one diameter signal of diameter versus length of the fiber-containing structure; (3) optionally signal processing the diameter signal to obtain at least one signal-processed diameter signal; and (4) comparing the diameter signal, the signal-processed diameter signal, or combinations thereof, against at least one reference signal to generate at least one surface defect signal of a surface defect output versus the length of the fiber-containing structure.

[0049] According to the invention,

the defect detector measures the cross-sectional diameter of the fiber-containing structure by transmitting light onto the fiber-containing structure with a projector, detecting a silhouette image of the fiber-containing structure with an optical receiver, and calculating the cross-sectional diameter based on a reduction in an amount of the light detected by the optical receiver relative to a total amount of the

light transmitted by the projector.

[0050] **FIG. 3** illustrates one example of a defect detector **100** configured to measure the cross-sectional diameter of a fiber-containing structure in linear motion by detecting a silhouette image **105** of the fiber-containing structure **110**. In this embodiment the defect detector **100** includes a projector **115** containing a light source **120** and a lens **125**, as well as an optical receiver **130** for detecting the silhouette image **105**. The cross-sectional diameter of the fiber-containing structure **110** is calculated based on a reduction in an amount of light **135** detected by the optical receiver relative to the total amount of light **140** transmitted by the projector. The optical receiver **130** produces a diameter signal **145** that is sent to a processor **150** configured to (i) optionally perform signal processing of the diameter signal **145** to obtain a signal-processed diameter signal, and to (ii) compare the diameter signal **145**, the optional signal-processed diameter signal, or combinations thereof, against at least one reference signal **155** to general at least one surface defect signal **160** of surface defect output versus the length of the fiber-containing structure **110**. In the embodiment of **FIG. 3**, the optical receiver **130** includes an active-pixel sensor **165**.

[0051] Various projectors and optical detectors known in the relevant art may be used in defect detectors of the present disclosure. For example, the projector may include laser diodes and/or light-emitting diodes, and the optical receiver may include an active-pixel sensor. Active-pixel sensors that are used in optical detectors of the present disclosure may include photodiode image sensors, charge-coupled device (CCD) image sensors, complementary metal-oxide-semiconductor (CMOS) image sensors, or combinations thereof.

[0052] Methods of the present disclosure may employ commercially-available devices capable of detecting the silhouette image of a fiber-containing structure in linear motion. For example, transmissive dimension measuring devices described in US 2010/0271638 by Torii et al. may be used as defect detectors in methods and apparatuses of the present disclosure.

[0053] Methods of the present disclosure may also employ different types of detectors known in the relevant art that are capable of imaging the surface of a fiber-containing structure in linear motion. For example, the detection method may also include a step of

imaging a surface of the fiber-containing structure with at least one imaging detector capable of imaging the surface by illuminating the fiber-containing structure with an imaging light and receiving a reflected image of the surface with an imaging receiver. Alternatively, detection methods of the present disclosure may not include imaging a surface of the fiber-containing structure with an imaging detector that receives a reflected image of the surface.

[0054] **FIGs. 4A** and **4B** illustrate how defect detectors of the present disclosure can measure a cross-sectional diameter of a fiber-containing structure.

[0055] As explained and illustrated above, the defect detector **100** measures the cross-sectional diameter of the fiber-containing structure **110** by transmitting light **140** onto the fiber-containing structure **110** with a projector **115**, and then detecting a silhouette image **105** of the fiber-containing structure **110** with an optical receiver **130**, see **FIG. 3**.

[0056] **FIG. 4A** illustrates an embodiment wherein the projector **115** includes a light-projecting slit **170** through which light transmitted from the light source **120** passes and is shaped into a belt-shaped, parallel light beam **175**. **FIG. 4B** illustrates an embodiment wherein the corresponding optical receiver **130** includes a light-receiving slit **185** through which the partially-blocked parallel light beam **135** passes prior to being detected by, for example, the active-pixel sensor **165**.

[0057] In the illustrations of **FIGs. 4A** and **4B**, the presence of a defect-free portion **190** of the fiber-containing structure **110** (being linearly passed through the defect detector **100** at a linear rate ***v***) causes a portion of the parallel light beam **175** to be blocked—such that an amount $\mathbf{A}^{D}$ of the partially-blocked parallel light beam **135** detected by the optical receiver **130** is less than an amount $\mathbf{A}^{T}$ of the parallel light beam **175** transmitted by the projector **115**. The blocked portion of the parallel light beam **135** corresponds to the silhouette image **105** of the fiber-containing structure **110**. In this manner the cross-sectional diameter **D** of the fiber-containing structure **110** may be calculated based on the difference between the amount $\mathbf{A}^{T}$ of the transmitted parallel light beam **175** and the amount $\mathbf{A}^{D}$ of the partially-blocked parallel light beam **135**—*i.e.*, $\mathbf{D} \sim (\mathbf{A}^{T} - \mathbf{A}^{D})$.

[0058] **FIGs. 5A** and **5B** illustrate how the silhouette image **105** of the fiber-containing

structure changes when the defect detector **100** detects a defect.

[0059] FIG. 5A illustrates the light-receiving slit **185** of the defect detector **100** at a time t_1 when the parallel light beam **175** is partially blocked by the defect-free portion **190** of the fiber-containing structure **110** (being linearly passed through the defect detector **100** at a linear rate v). As explained above, the blocked portion of the parallel light beam **135** corresponds to a silhouette image **105´** of the fiber-containing structure **110**. In this manner the cross-sectional diameter D_1 of the defect-free portion **190** at the time t_1 may be calculated based on the difference between the amount A^T of the transmitted parallel light beam **175** and the amount $A^D{}_1$ of the partially-blocked parallel light beam **135**—*i.e.*, $D_1 \sim (A^T - A^D{}_1)$.

[0060] FIG. 5B illustrates the light-receiving slit **185** of the defect detector **100** at a time t_2 when the parallel light beam **175** is partially blocked by a defect-containing portion **195** of the fiber-containing structure **110** (being linearly passed through the defect detector **100** at a linear rate v). At time t_2, the blocked portion of the parallel light beam **200** corresponds to a silhouette image **105´´** of the fiber-containing structure **110**. In this manner the cross-sectional diameter D_2 of the defect-containing portion **195** at the time t_2 may be calculated based on the difference between the amount A^T of the transmitted parallel light beam **175** and the amount $A^D{}_2$ of the partially-blocked parallel light beam **195**—*i.e.*, $D_2 \sim (A^T - A^D{}_2)$.

[0061] Because the defect-containing portion **195** illustrated in **FIG. 5B** includes the defect **67** in the form of a "fluff" or "peel" caused by a broken filament, more of the parallel light beam **175** is blocked at the time t_2 compared to the time t_1. Consequently, in this illustration, the calculated cross-sectional diameter D_2 of the defect-containing portion **195** in **FIG. 5B** is greater than the calculated cross-sectional diameter D_1 of the defect-free portion **190** in **FIG. 5A**. In other embodiments, the cross-sectional diameter D_2 of the defect-containing portion may be less than the cross-sectional diameter D_1 of the defect- free portion of the fiber-containing structure **110**. For example, when a portion of a broken filament breaks off from the fiber-containing structure, then the resulting defect-containing portion may have a smaller cross-sectional diameter compared to the

corresponding cross-sectional diameter of the defect-free portions of the fiber-containing structure.

[0062] As explained below in greater detail, the measuring of at least one cross-sectional diameter of the fiber-containing structure with at least one defect detector occurs at regular (time / distance) intervals as the fiber-containing structure is linearly passed through the defect detector at a linear rate $\mathbf{v}$. In doing so, the measuring of the at least one cross-sectional diameter may occur at a sampling rate ranging from about 1 sample-per-second to at least 5,000 samples-per-second. In some embodiments the sampling rate of at least one of the defect detector is adjusted such that the measuring occurs at constant intervals along the length of the fiber-containing structure. Constant intervals over which the measurements occur may range from about 10 nm to about 1 cm.

[0063] Following the steps of measuring at least one cross-sectional diameter of the fiber-containing structure with at least one defect detector, and optionally signal processing the diameter signal to obtain at least one signal-processed diameter signal, methods of the present disclosure may include the step of comparing the diameter signal and/or the signal-processed diameter signal against at least one reference signal to generate at least one surface defect signal of a surface defect output versus the length of the fiber-containing structure. As explained below, the at least one surface defect signal may relate to the magnitude, slope and/or curvature of the diameter signal.

[0064] The surface defect signal may include a magnitude surface defect count signal of a magnitude surface defect count versus the length of the fiber-containing structure. The magnitude surface defect count signal is generated by comparing the diameter signal against a reference signal of a maximum cross-sectional diameter versus the length of the fiber-containing structure. A magnitude surface defect count of zero occurs when a magnitude of the diameter signal is less than or equal to the maximum cross-sectional diameter at a particular point along the length of the fiber-containing structure. A magnitude surface defect count of greater than zero occurs when the magnitude of the diameter signal is greater than the maximum cross-sectional diameter at the particular point along the length of the fiber-containing structure, such that the magnitude surface defect count of greater than zero may be a positive integer corresponding to a percentage

of the magnitude of the diameter signal above the maximum cross-sectional diameter relative.

[0065] The surface defect signal according to the invention includes a slope surface defect count signal of a

slope surface defect count versus the length of the fiber-containing structure. The slope surface defect count signal is generated by comparing the signal-processed diameter signal against a reference signal of a maximum first derivative of the diameter signal versus the length of the fiber-containing structure, where the signal-processed diameter signal is obtained by signal processing the diameter signal to obtain a first derivative of the diameter signal. A slope surface defect count of zero occurs when an absolute value of the first derivative of the diameter signal is less than or equal to the maximum first derivative of the diameter signal at a particular point along the length of the fiber-containing structure. A slope surface defect count of greater than zero occurs when the absolute value of the first derivative of the diameter signal is greater than the maximum first derivative of the diameter signal at the particular point along the length of the fiber-containing structure, such that the slope surface defect count of greater than zero may be a positive integer corresponding to a percentage of the absolute value of the first derivative of the diameter signal above the maximum first derivative of the diameter signal.

[0066] The surface defect signal according to the invention includes a curvature surface defect count signal of

a curvature surface defect count versus the length of the fiber-containing structure. The curvature surface defect count signal is generated by comparing the signal- processed diameter signal against a reference signal of a maximum second derivative of the diameter signal versus the length of the fiber-containing structure, where the signal-processed diameter signal is obtained by signal processing the diameter signal to obtain a second derivative of the diameter signal. A curvature surface defect count of zero occurs when an absolute value of the second derivative of the diameter signal is less than or equal to the maximum second derivative of the diameter signal at a particular point along the length of the fiber-containing structure. A curvature surface defect count of greater than zero occurs when the absolute value of the second derivative of the diameter signal is greater than the maximum second derivative of the diameter signal at the particular point along the length of the fiber-containing structure, such that the curvature

surface defect count of greater than zero may be a positive integer corresponding to a percentage of the absolute value of the second derivative of the diameter signal above the maximum second derivative of the diameter signal.

[0067] Detection methods of the present disclosure may also include a step of differentiating and/or categorizing defects contained in a fiber-containing structure based, at least in part, on the magnitude, slope and/or curvature of the diameter signal. Detection methods of the present disclosure may also include a step of generating a surface defect rating of the fiber-containing structure based on the magnitude, slope and/or curvature of the diameter signal.

[0068] FIGs. 6A thru **6C** illustrate how changes in the cross-sectional diameters of defect-containing portions of fiber-containing structures can be used to count, differentiate and categorize different defects.

[0069] FIG. 6A illustrates the location of cross-sectional measurements taken of a fiber-containing structure **110** at regular intervals occurring at times t_1 through t_8. The fiber-containing structure **110** (being linearly passed through the defect detector **100** at a linear rate v) includes a defect-containing portion **195** having a defect **65** in the form of a "fluff" or a "peel" caused by a broken filament. In this illustration, the calculated cross-sectional diameters at times t_2, t_3, t_4 and t_5 indicate the presence of the "fluff" or "peel" defect **65** caused by the broken filament.

[0070] Focusing on the magnitude of cross-sectional diameter of the fiber-containing structure **110** within the defect-containing portion **195**, **FIG. 6A** illustrates wherein (i) the magnitude slightly increases at the time t_3 relative to the time t_2 (magnitude surface defect count of slightly greater than zero occurs at the time t_3), (ii) the magnitude increases more dramatically at time t_4 relative to time t_2 (magnitude surface defect count of significantly greater than zero occurs at time t_4), and then (iii) the magnitude abruptly decreases at the time t_5 relative to the time t_4 (magnitude surface defect count of zero occurs at time t_5). This pattern of the slight increase in the magnitude surface defect count at the time t_3, in conjunction with the abrupt decrease in the magnitude surface defect count at the time t_5 back to zero, indicates the presence of a "fluff" or "peel" defect caused by a broken

filament having only a single branch (*i.e.*, a trailing-edge branch extending outward at an acute angle relative to the direction of travel – *see* **FIG. 2A**).

[0071] Focusing on the slope of cross-sectional diameter of the fiber-containing structure **110** within the defect-containing portion **195**, **FIG. 6A** illustrates wherein (i) the slope increases at the time t_3 relative to the time t_2 (slope surface defect count of slightly greater than zero occurs at the time t_3), (ii) the slope remains constant or slightly increases at time t_4 relative to time t_3 (slope surface defect count of greater than zero occurs at time t_4), and then (iii) the slope abruptly decreases at the time t_5 relative to the time t_4 (slope surface defect count of zero occurs at time t_5). This pattern of the increase in the slope surface defect count at the time t_3, in conjunction with the abrupt decrease in the slope surface defect count at the time t_5 back to zero, indicates the presence of a "fluff" or "peel" defect caused by a broken filament having only a single branch (*i.e.*, a trailing-edge branch extending outward at an acute angle relative to the direction of travel – *see* **FIG. 2A**).

[0072] **FIG. 6B** illustrates the location of cross-sectional measurements taken of a fiber-containing structure **110** at regular intervals occurring at times t_1 through t_8. The fiber-containing structure **110** (being linearly passed through the defect detector **100** at a linear rate v) includes a defect-containing portion **195** having a defect **85** in the form of a small "pill' or a "slub" caused by an impurity attached to the surface of the fiber-containing structure **110**. In this illustration, the calculated cross-sectional diameters at times t_2, t_3, t_4 and t_5 indicate the presence of the "pill" or "slub" defect **85** caused by the impurity.

[0073] Focusing on the magnitude of cross-sectional diameter of the fiber-containing structure **110** within the defect-containing portion **195**, **FIG. 6B** illustrates wherein (i) the magnitude abruptly increases at the time t_3 relative to the time t_2 (magnitude surface defect count of significantly greater than zero occurs at the time t_3), (ii) the magnitude increases only slightly at time t_4 relative to time t_3 (magnitude surface defect count increases slightly at time t_4 relative to the time t_3), and then (iii) the magnitude abruptly decreases at the time t_5 relative to the time t_4 (magnitude surface defect count of zero occurs at time t_5). This pattern of the abrupt and significant increase in the magnitude surface defect count at the time t_3, in conjunction with the abrupt decrease in the magnitude surface defect count at the time t_5 back to zero, indicates the presence of a

small "pill" or "slub" defect caused by a recently-introduced impurity.

[0074] Focusing on the slope of cross-sectional diameter of the fiber-containing structure **110** within the defect-containing portion **195**, **FIG. 6B** illustrates wherein (i) the slope increases at the time t_3 relative to the time t_2 (slope surface defect count of greater than zero occurs at the time t_3), (ii) the slope reduces to nearly zero at the time t_4 relative to the time t_3 (slope surface defect count of around zero occurs at time t_4), and then (iii) the slope returns to zero at the time t_5 relative to the time t_4 (slope surface defect count of zero occurs at time t_5). This pattern of the increase in the slope surface defect count at the time t_3, in conjunction with the abrupt decrease in the slope surface defect count at the time t_4 and the decrease in the slope surface defect count back to zero at the time t_5, indicates the presence of a "pill" or "slub" defect caused by a recently-introduced impurity.

[0075] Focusing on the curvature of cross-sectional diameter of the fiber-containing structure **110** within the defect-containing portion **195**, **FIG. 6B** illustrates wherein (i) the curvature increases at the time t_3 relative to the time t_2 (curvature surface defect count of greater than zero occurs at the time t_3), (ii) the curvature reduces to nearly zero at the time t_4 relative to the time t_3 (curvature surface defect count of near zero occurs at time t_4), and then (iii) the curvature remains at zero at the time t_5 relative to the time t_4 (curvature surface defect count of zero occurs at time t_5). This pattern of the increase in the curvature surface defect count at the time t_3, in conjunction with the reduction in curvature surface defect count to zero at the times t_4 and t_5, indicates the presence of a "pill" or "slub" defect caused by a recently-introduced impurity.

[0076] FIG. 6C illustrates the location of cross-sectional measurements taken of a fiber-containing structure **110** at regular intervals occurring at times t_1 through t_8. The fiber-containing structure **110** (being linearly passed through the defect detector **100** at a linear rate v) includes a defect-containing portion **195** having a defect **95** in the form of an elongated or enlarged "pill' or a "slub" caused by the built-up or shaping of impurities attached to the surface of the fiber-containing structure **110**. In this illustration, the calculated cross-sectional diameters at times t_2, t_3, t_4 and t_5 indicate the presence of the "pill" or "slub" defect **95** caused by the impurity.

[0077] Focusing on the magnitude of cross-sectional diameter of the fiber-containing structure **110** within the defect-containing portion **195**, **FIG. 6C** illustrates wherein (i) the magnitude increases slightly at the time t_3 relative to the time t_2 (magnitude surface defect count of slightly greater than zero occurs at the time t_3), (ii) the magnitude increases more dramatically at time t_4 relative to time t_3 (magnitude surface defect count increases dramatically at time t_4 relative to the time t_3), and then (iii) the magnitude decreases at the time t_5 relative to the time t_4 (magnitude surface defect count of slightly greater than zero occurs at time t_5). This pattern of the slight increase in the magnitude surface defect count at the time t_3, in conjunction with the increase in the magnitude surface defect count at the time t_4 and then the decrease in the magnitude surface defect count at the time t_5, indicates the presence of an elongated or enlarged "pill' or a "slub" caused by the built-up or shaping of impurities.

[0078] Focusing on the slope of cross-sectional diameter of the fiber-containing structure **110** within the defect-containing portion **195**, **FIG. 6C** illustrates wherein (i) the slope increases at the time t_3 relative to the time t_2 (slope surface defect count of greater than zero occurs at the time t_3), (ii) the slope reduces to nearly zero at the time t_4 relative to the time t_3 (slope surface defect count of around zero occurs at time t_4), and then (iii) the slope increases again at the time t_5 relative to the time t_4 (slope surface defect count of greater than zero occurs at time t_5). This pattern of the increase in the slope surface defect count at the time t_3, in conjunction with the abrupt decrease in the slope surface defect count to around zero at the time t_4 and the increase in the slope surface defect count at the time t_5, indicates the presence of an elongated or enlarged "pill' or a "slub" caused by the built-up or shaping of impurities.

[0079] Focusing on the curvature of cross-sectional diameter of the fiber-containing structure **110** within the defect-containing portion **195**, **FIG. 6C** illustrates wherein (i) the curvature increases at the time t_3 relative to the time t_2 (curvature surface defect count of greater than zero occurs at the time t_3), (ii) the curvature reduces to nearly zero at the time t_4 relative to the time t_3 (curvature surface defect count of around zero occurs at time t_4), and then (iii) the curvature increases again at the time t_5 relative to the time t_4 (curvature surface defect count of greater than zero occurs at time t_5). This pattern of the increase in the curvature surface defect count at the time t_3, in conjunction with the abrupt

decrease in the curvature surface defect count to around zero at the time t_4 and the increase in the curvature surface defect count at the time t_5, indicates the presence of an elongated or enlarged "pill' or a "slub" caused by the built-up or shaping of impurities.

[0080] In general, the ability to differentiate and categorize different defects contained in a fiber-containing structure, being linearly passed through at least one defect detector, can be improved by increasing the sampling rate of the measuring. Thus, for example, if the number of the measurement intervals in **FIGs. 6A** thru **6C** were doubled—such that 16 measurements are taken over the same length of the fiber-containing structure **110**—then the ability to differentiate and categorize the different defects could be improved, because the patterns described above could be detected with greater resolution.

[0081] In some embodiments the defect detection method may include an additional step of changing a linear rate of the fiber-containing structure, based on the surface defect signal. For example, the linear rate of the fiber-containing structure may be reduced in order to increase the number of measurement intervals—thereby increasing the sensitivity and improving the ability to differentiate and categorize the different defects. In this regard, defect detection methods of the present disclosure may employ at least one speedometer to measure the linear rate of the fiber-containing structure. In some embodiments the fiber-containing structure may be linearly passed through the at least one defect detector at a linear rate of at least 300 meters per minute, while in other embodiments the linear rate may be less than 10 centimeters per minute. In some embodiments, for example where the fiber-containing structure is a braided fiber-containing structure or a wire-laid fiber-containing structure having textured surfaces, the fiber-containing structure may be linearly passed through the at least one detector at a linear rate of from less than 10 centimeters per minute to at least 1 meter per minute.

[0082] In some embodiments the defect detection method may include an additional step of changing the sampling rate of measuring the at least one cross-sectional diameter, based on the surface defect signal. For example, the sampling rate may be increased in order to increase the number of measurement intervals—thereby increasing the sensitivity and improving the ability to differentiate and categorize the different defects.

[0083] Regarding the reference signals used to generate the various surface defect

signals (e.g., based on the magnitude, slope and/or curvature of the cross-sectional diameter), the reference signals may be constant reference signals that are constant along the length of the fiber-containing structure, the reference signals may be variable reference signals that change at one or more points along the length of the fiber-containing structure, or combinations thereof.

[0084] At least one of the defect detector may be situated in series with at least one roller, at least one extrusion apparatus configured to form the fiber-containing structure, at least one braiding machine configured to form the fiber-containing structure, at least one tensioning assembly configured to apply tension to the fiber-containing structure, at least one finish applicator configured to apply a coating to the fiber-containing structure, at least one godet roll assembly configured to stretch the fiber-containing structure, at least one winding assembly configured to wind the fiber-containing structure onto a bobbin, or combinations thereof. Other apparatuses commonly used to handle, process and/or measure fiber-containing structure may also be situated in series with defect detectors of the present disclosure.

[0085] FIG. 7 illustrates an apparatus for producing or processing a fiber-containing structure, in which a plurality of defect detectors **205, 206, 207, 208, 209 and 210** (or the defect detector arrays described below) are situated with a tensioning assembly **215**, a finish applicator **220**, a godet roll assembly **225**, and a winding assembly **230**. In this example, the fiber-containing structure **110** is linearly passed from a bobbin or upstream process **235** thru the tensioning assembly **215**, which is situated in series with the surrounding defect detectors **205** and **206**, then over a first roller **240** and thru the finish applicator **220**, which is situated in series with the surrounding defect detectors **207** and **208**, then over a second roller **245** and thru the godet roll assembly **225**, which is situated in series with the defect detector **209**, then over a third roller **250** and into the winding assembly **230** which includes a dancer arm **255**, a traverse guide assembly **260** and a finish bobbin **265**. Upstream processes **235** may include, for example, an extrusion, winding or braiding process for producing the fiber-containing structure **110**.

[0086] Detection methods of the present disclosure may include a step of forming the fiber-containing structure by an extrusion process, wherein at least one of the defect detector is situated in series with the extrusion apparatus. Detection methods of the

present disclosure may also include a step of forming the fiber-containing structure by a braiding process, wherein at least one of the defect detector is situated in series with the braiding machine. Defect detectors may also be situated within a braiding apparatus, for example, between at least one carrier bobbin (or carrier guide) and the winding shaft.

[0087] Detection methods of the present disclosure may also include a step of modifying an operation of the extrusion apparatus, the braiding machine, the tensioning assembly, the finish applicator, the godet roll assembly, the winding assembly, the linear rate, or combinations thereof, based on the surface defect signal. For example, if the method detects an increase in the surface defect output beginning immediately downstream of the finish applicator **220**, then the operation of the finish applicator could be modified by reducing the linear rate v of the fiber-containing structure, or by altering the operational parameters of the finish applicator **220** to reduce the surface defect output.

[0088] Defect detector arrays comprising at least two defect detector situated in series may also be employed in methods of the present disclosure. **FIG. 8** illustrates an embodiment wherein a defect detector array **270** is situated downstream of a braiding machine **275** and upstream of a downstream process **280** such as the tensioning assembly **215**, the finish applicator **220**, the godet roll assembly **225**, and/or the winding assembly **230** described above. In the example of **FIG 8**, the defect detector array **270** includes two defect detectors **285** and **290** situated in series. In some embodiments the defect detectors of a defect detector array may be rotated relative to one another by a detector offset angle, such that different silhouette images of the fiber-containing structure are detected by the at least two detect detectors. The detector offset angle may range from 1° to less than 360°.

[0089] Defect detectors of the present disclosure may include multi-axis defect detectors capable of simultaneously measuring a plurality of cross-sectional diameters of a fiber-containing structure from different directions.

[0090] In one embodiment a biaxial defect detector includes two projectors for transmitting the light onto the fiber-containing structure, and two optical receivers for detecting the silhouette image of the fiber-containing structure. The two projectors may

include a projector A for transmitting a light A onto a surface A of the fiber-containing structure, and a projector B for transmitting a light B onto a surface B of the fiber-containing structure. The two optical receivers may include an optical receiver A for detecting a silhouette image A of the fiber-containing structure, and an optical receiver B for detecting a silhouette image B of the fiber-containing structure. In this scenario, the projector A is optically aligned with the optical receiver A, and the projector B is optically aligned with the optical receiver B. The projector A and the projector B are rotated relative to one another by a projector offset angle, and the optical receiver A and the optical receiver B are rotated relative to one another by an optical receiver offset angle. In some embodiments the projector offset angle and the optical receiver offset angle may independently range from 5° to 175°.

[0091] FIG. 9 illustrates a defect detector array **295** including two biaxial defect detectors α **300** and β **305** mounted in series relative to a fiber-containing structure **110** moving linearly through the defect detector array **295** at a linear rate v. Each of the two biaxial defect detectors α **300** and β **305** include the projector A **310** and the projector B **315**, as well as the optical receiver A **320** and the optical receiver B **325**. The projector A **310** is optically aligned with the optical receiver A **320**, and the projector B **315** is optically aligned with the optical receiver B **325**. In the biaxial defect detector α **300**, the projector A **310** and the projector B **315** are rotated relative to one another by a projector offset angle $\Delta\Phi_{\alpha}$, where $\Delta\Phi_{\alpha} = \Phi^{1}_{\alpha} - \Phi^{2}_{\alpha}$. In the biaxial defect detector β **305**, the projector A **310** and the projector B **315** are rotated relative to one another by a projector offset angle $\Delta\Phi_{\beta}$, where $\Delta\Phi_{\beta} = \Phi^{1}_{\beta} - \Phi^{2}_{\beta}$.

[0092] In the embodiment of **FIG. 9** the biaxial defect detectors α **300** and β **305** are also offset relative to one another by a detector offset angle of approximately 90°, and set apart by a linear separation distance d **330**. In some embodiments the detector offset angle may range from 1° to less than 360°. In some embodiments the separation distance d may range from about 1 mm to about 100 mm. Other embodiments of the present disclosure may employ defect detector arrays having at least three of the defect detectors arranged in series.

Apparatuses for Detecting Defects in Moving Fiber-Containing Structures

[0159] The present disclosure also relates to apparatuses for detecting defects in moving fiber-containing structures. Such apparatus may include (A) the extrusion apparatus configured to form a fiber-containing structure, the braiding machine configured to form the fiber-containing structure, the tensioning assembly configured to apply tension to the fiber-containing structure, the finish applicator configured to apply a coating to the fiber-containing structure, the godet roll assembly configured to stretch the fiber-containing structure, the winding assembly configured to wind the fiber-containing structure onto a bobbin, the roller, or combinations thereof; (B) a defect detector configured to measure at least one cross-sectional diameter of the fiber-containing structure by transmitting light onto the fiber-containing structure with a projector, detecting a silhouette image of the fiber-containing structure with an optical receiver, and calculating the cross-sectional diameter based on a reduction in an amount of the light

detected by the optical receiver relative to a total amount of the light transmitted by the projector; and (C) a processor configured to compare at least one diameter signal obtained from the defect detector, optionally at least one signal-processed diameter signal, or combinations thereof, against at least one reference signal to obtain at least one surface defect signal of a surface defect output versus length of the fiber-containing structure, wherein the defect detector measures the at least one cross-sectional diameter while the fiber-containing structure is linearly passing through the defect detector.

[0160] Apparatuses of the present disclosure are beyond the claimed invention, operate as described in the methods

above, and may be modified in accordance with the subject matter described above.

CLAIMS

What is claimed is:

1. A method, comprising:

linearly passing a fiber-containing structure through at least one defect detector;

measuring at least one cross-sectional diameter of the fiber- containing structure with the defect detector to obtain at least one diameter signal of diameter versus length of the fiber-containing structure;

signal processing the diameter signal to obtain at least one signal-processed diameter signal; and

comparingthe signal-processed diameter signal against at least one reference signal to generate at least one surface defect signal of a surface defect output versus the length of the fiber-containing structure,

wherein:

the defect detector measures the cross-sectional diameter of the fiber-containing structure by transmitting light onto the fiber-containing structure with a projector, detecting a silhouette image of the fiber-containing structure with an optical receiver, and calculating the cross-sectional diameter based on a reduction in an amount of the light detected by the optical receiver relative to a total amount of the light transmitted by the projector; and

at least one of the defect detector is situated in series with an extrusion apparatus configured to form the fiber-containing structure, a braiding machine configured to form the fiber-containing structure, a tensioning assembly configured to apply tension to the fiber-containing structure, a finish applicator configured to apply a coating to the fiber-containing structure, a godet roll assembly configured to stretch the fiber-containing structure, a winding assembly configured to wind the fiber-containing structure onto a bobbin, or combinations thereof,

characterized in that the at least one surface defect signal comprises a slope surface defect count signal of a slope surface defect count versus the length of the fiber-containing structure, in which:

the slope surface defect count signal is generated by comparing the signal-processed diameter signal against a reference signal of a maximum first derivative of the diameter signal versus the length of the fiber-containing structure, where the signal-processed diameter signal is obtained by signal processing the diameter signal to obtain a first derivative of the diameter signal;

a slope surface defect count of zero occurs when an absolute value of the first derivative of the diameter signal is less than or equal to the maximum first derivative of the diameter signal at a particular point along the length of the fiber-containing structure; and

a slope surface defect count of greater than zero occurs when the absolute value of the first derivative of the diameter signal is greater than the maximum first derivative of the diameter signal at the particular point along the length of the fiber-containing structure, such that the slope surface defect count of greater than zero is a positive integer corresponding to a percentage of the absolute value of the first derivative of the diameter signal above the maximum first derivative of the diameter signal.

2. A method, comprising:

linearly passing a fiber-containing structure through at least one defect detector;

measuring at least one cross-sectional diameter of the fiber- containing structure with the defect detector to obtain at least one diameter signal of diameter versus length of the fiber-containing structure;

signal processing the diameter signal to obtain at least one signal-processed diameter signal; and

comparingthe signal-processed diameter signal against at least one reference signal to generate at least

one surface defect signal of a surface defect output versus the length of the fiber-containing structure,

wherein:

the defect detector measures the cross-sectional diameter of the fiber-containing structure by transmitting light onto the fiber-containing structure with a projector, detecting a silhouette image of the fiber-containing structure with an optical receiver, and calculating the cross-sectional diameter based on a reduction in an amount of the light detected by the optical receiver relative to a total amount of the light transmitted by the projector; and

at least one of the defect detector is situated in series with an extrusion apparatus configured to form the fiber-containing structure, a braiding machine configured to form the fiber-containing structure, a tensioning assembly configured to apply tension to the fiber-containing structure, a finish applicator configured to apply a coating to the fiber-containing structure, a godet roll assembly configured to stretch the fiber-containing structure, a winding assembly configured to wind the fiber-containing structure onto a bobbin, or combinations thereof,

characterized in that the at least one surface defect signal comprises a curvature surface defect count signal of a curvature surface defect count versus the length of the fiber-containing structure, in which:

the curvature surface defect count signal is generated by comparing the signal-processed diameter signal against a reference signal of a maximum second derivative of the diameter signal versus the length of the fiber-containing structure, where the signal-processed diameter signal is obtained by signal processing the diameter signal to obtain a second derivative of the diameter signal;

a curvature surface defect count of zero occurs when an absolute value of the second derivative of the diameter signal is less than or equal to the maximum second derivative of the diameter signal at a particular point along the length of the fiber-containing structure; and

a curvature surface defect count of greater than zero occurs when the absolute value of the second derivative of the diameter signal is greater than the maximum second derivative of the diameter signal at the particular point along the

length of the fiber-containing structure, such that the curvature surface defect count of greater than zero is a positive integer corresponding to a percentage of the absolute value of the second derivative of the diameter signal above the maximum second derivative of the diameter signal.

3. The method of claim 1 or 2, further comprising:

forming the fiber-containing structure by an extrusion process, wherein at least one of the defect detector is situated in series with the extrusion apparatus.

4. The method of claim 1 or 2, further comprising:

forming the fiber-containing structure by a braiding process, wherein at least one of the defect detector is situated in series with the braiding machine.

5. The method of claim 4, further comprising:

forming the fiber-containing structure by a braiding process,

wherein:

a defect detector array is situated in series with the braiding machine, such that the fiber-containing structure linearly passes through the defect detector array; the defect detector array comprises at least two of the defect detector situated in series;

the at least two of the defect detector are rotated relative to one another by a detector offset angle, such that different silhouette images of the fiber-containing structure are detected by the at least two of the detect detector; and

the detector offset angle ranges from 1° to less than 360°.

6. The method of claim 5, wherein the defect detector array comprises at least three of the defect detector situated in series.

7. The method of any one of claims 1-6, wherein:

the defect detector comprises two projectors for transmitting the light onto

the fiber-containing structure and two optical receivers for detecting the silhouette image of the fiber-containing structure;

the two projectors comprise a projector A, for transmitting a light A onto a surface A of the fiber-containing structure, and a projector B, for transmitting a light B onto a surface B of the fiber-containing structure;

the two optical receivers comprise an optical receiver A, for detecting a silhouette image A of the fiber-containing structure, and an optical receiver B, for detecting a silhouette image B of the fiber-containing structure;

the projector A is optically aligned with the optical receiver A, and the projector B is optically aligned with the optical receiver B;

the projector A and the projector B are rotated relative to one another by a projector offset angle, and the optical receiver A and the optical receiver B are rotated relative to one another by an optical receiver offset angle; and

the projector offset angle and the optical receiver offset angle range from 5° to 175°.

8. The method of any one of claims 1-7, wherein the defect detector is a multi-axis defect detector configured to measure at least two cross-sectional diameters of the fiber- containing structure, such that an offset angle between the at least two cross- sectional diameters ranges from 5° to 175°.

9. The method of any one of claims 1-8, further comprising:

imaging a surface of the fiber-containing structure with at least one imaging detector,

wherein:

the imaging detector images the surface by illuminating the fiber-containing structure with an imaging light, and receiving a reflected image of the surface with an imaging receiver; and

at least one of the imaging detector is situated in series with the extrusion apparatus, the braiding machine, the tensioning assembly, the finish applicator, the godet roll assembly, the winding assembly, or the combinations thereof.

10. The method of any one of claims 1-9, further comprising:

measuring a linear rate of the fiber-containing structure using at least one speedometer.

11. The method of any one of claims 1-9, further comprising:

modifying an operation of the extrusion apparatus, the braiding machine, the tensioning assembly, the finish applicator, the godet roll assembly, the winding assembly, or combinations thereof, based on the surface defect signal.

FIG. 1A

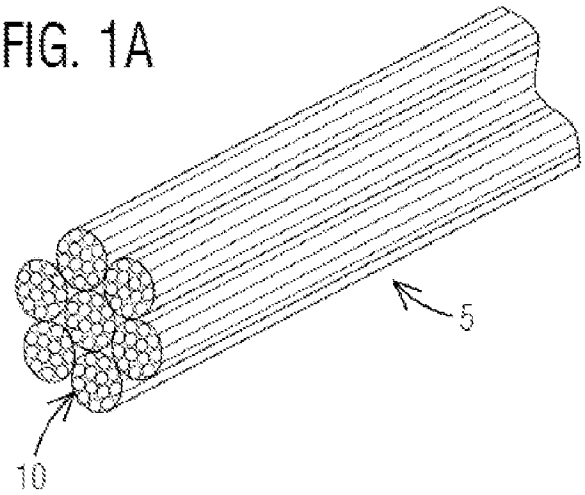


FIG. 1B

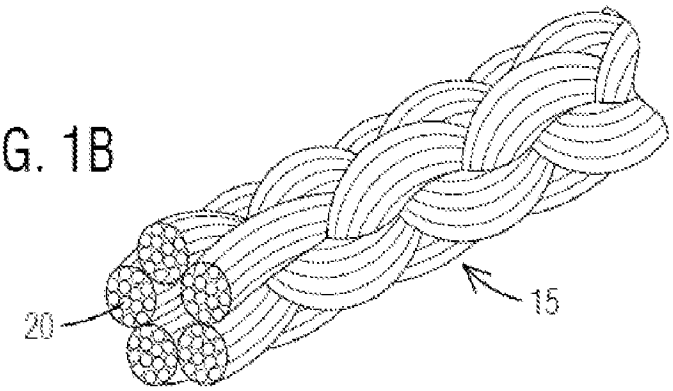


FIG. 1C

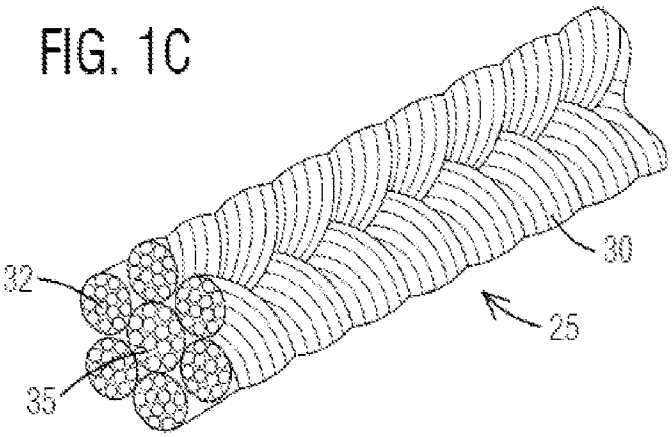


FIG. 1D

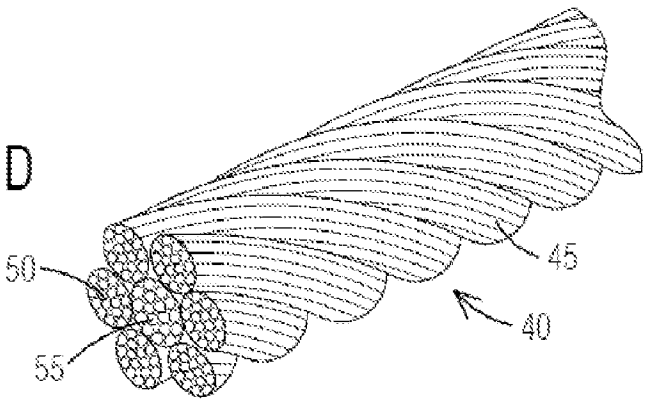


FIG. 2A

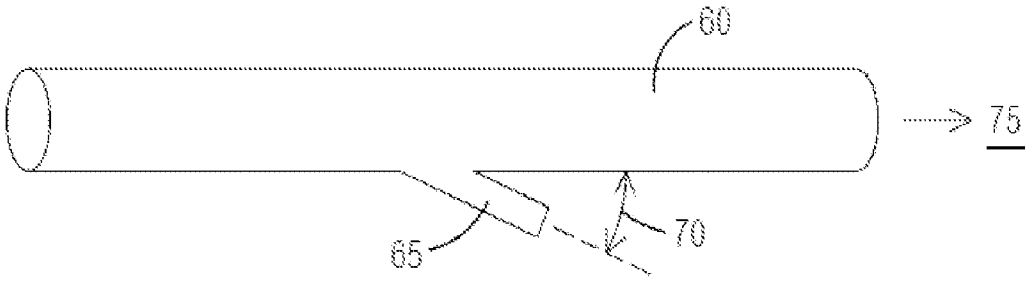


FIG. 2B

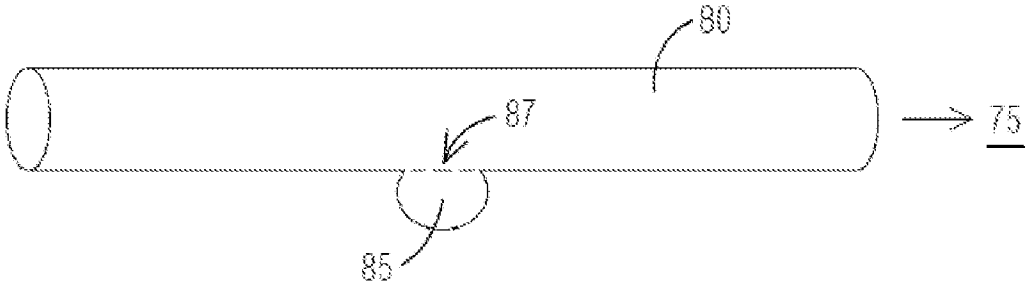


FIG. 2C

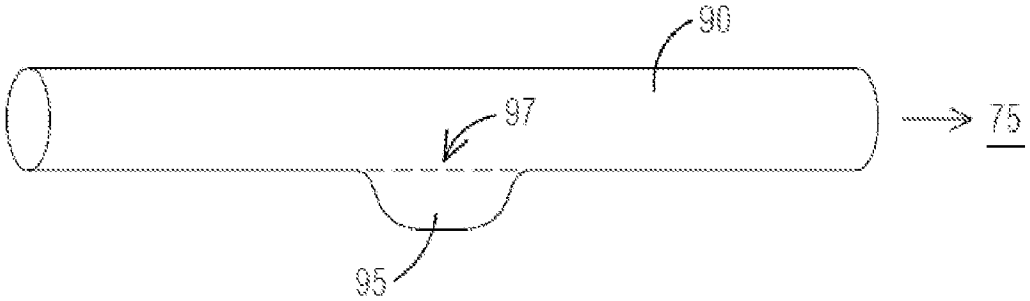


FIG. 3

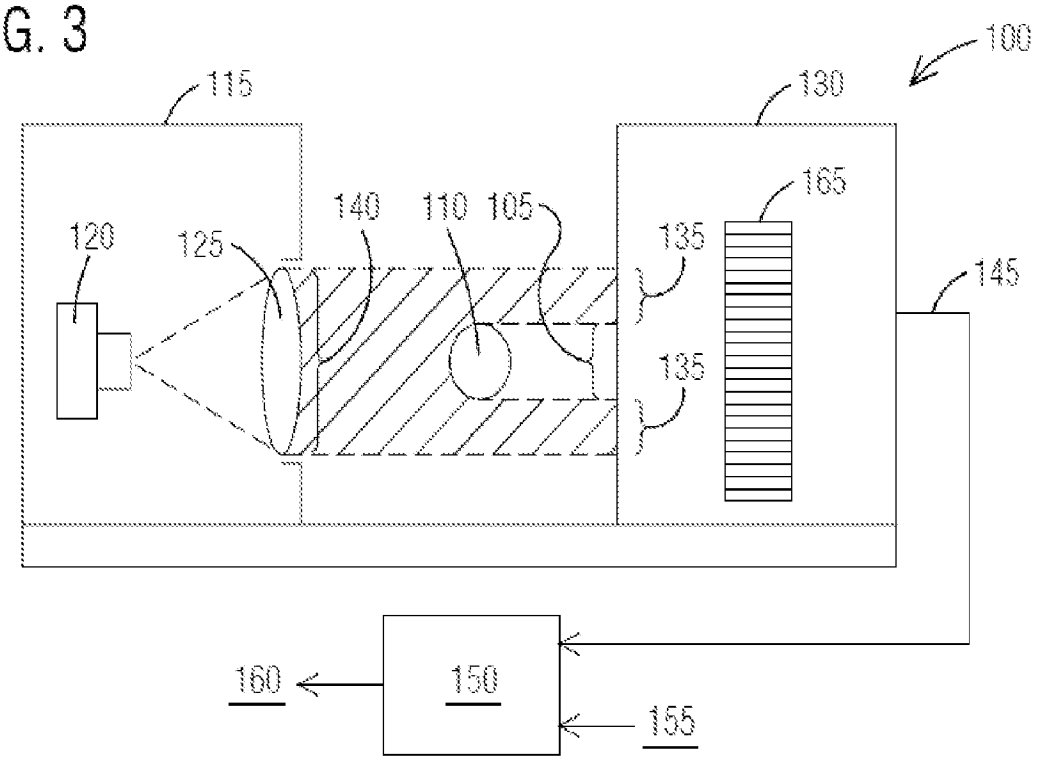


FIG. 4A

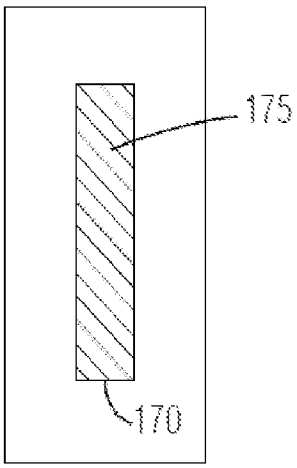


FIG. 4B

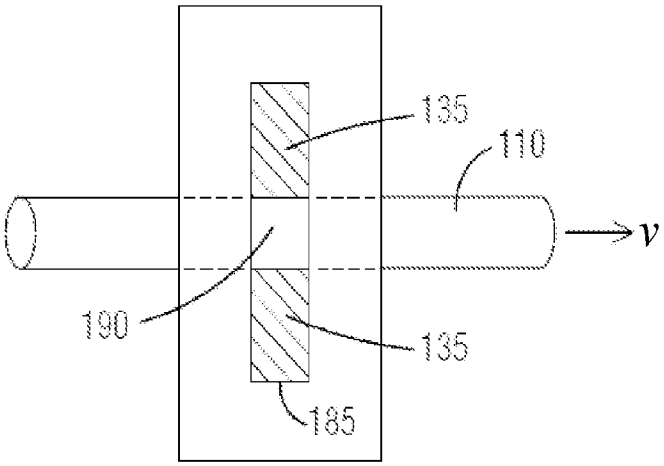


FIG. 5A

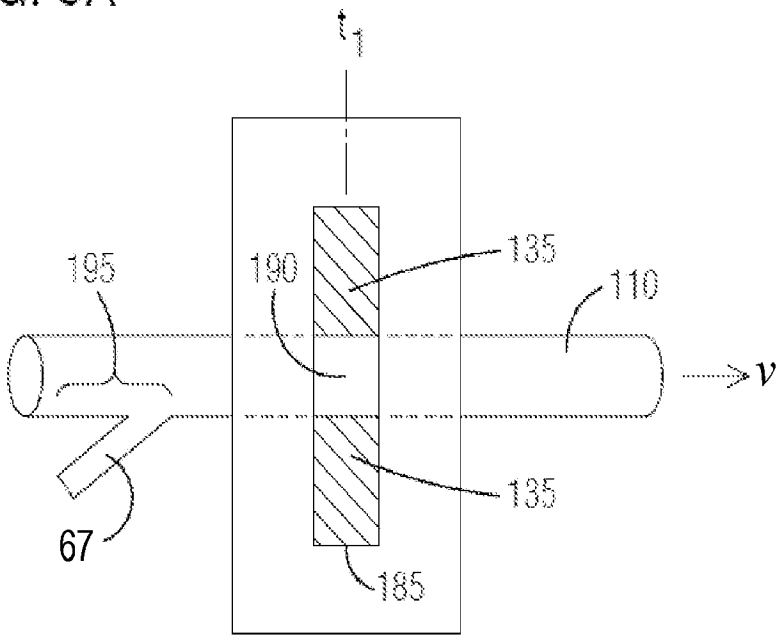


FIG. 5B

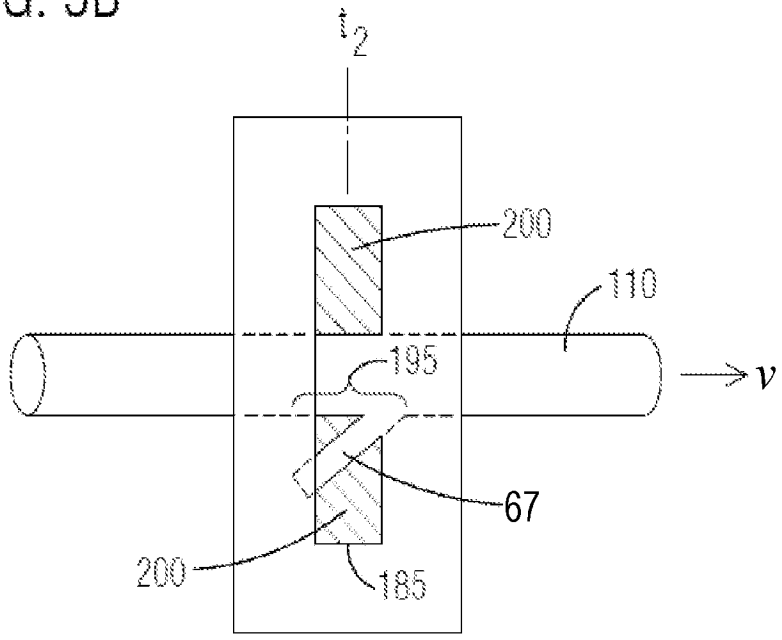


FIG. 6A

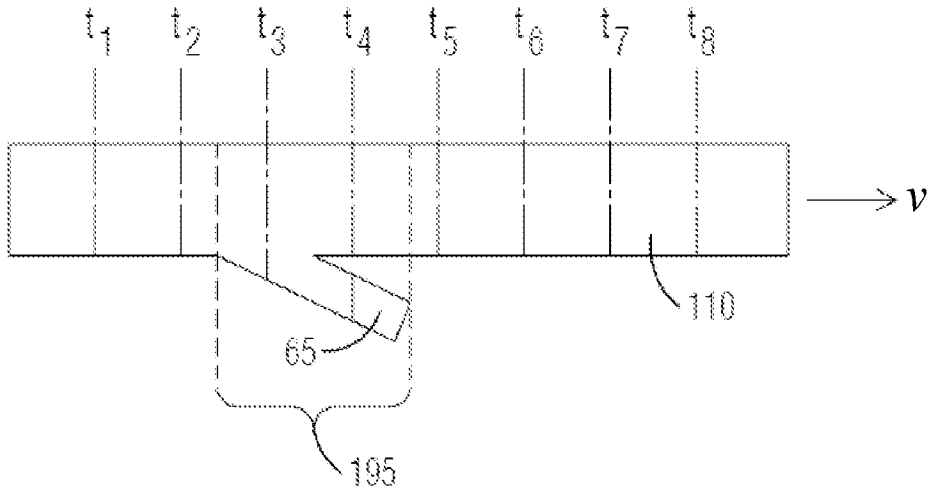


FIG. 6B

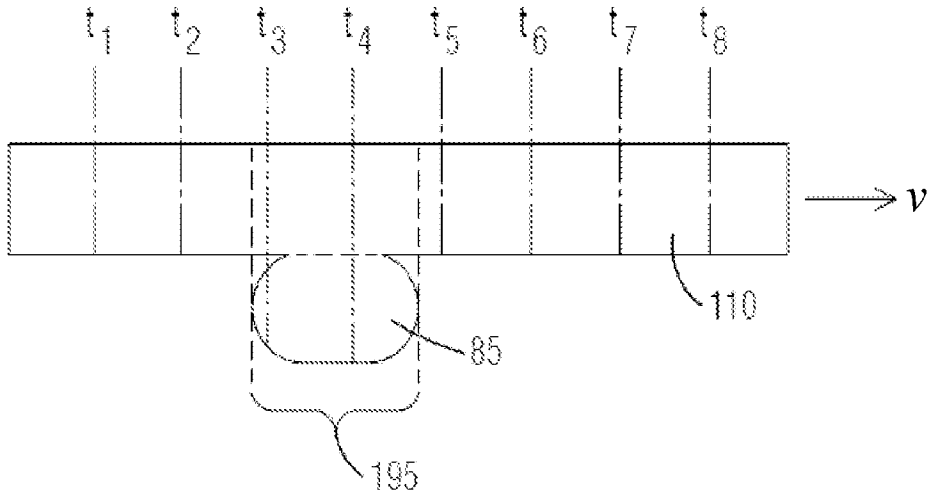


FIG. 6C

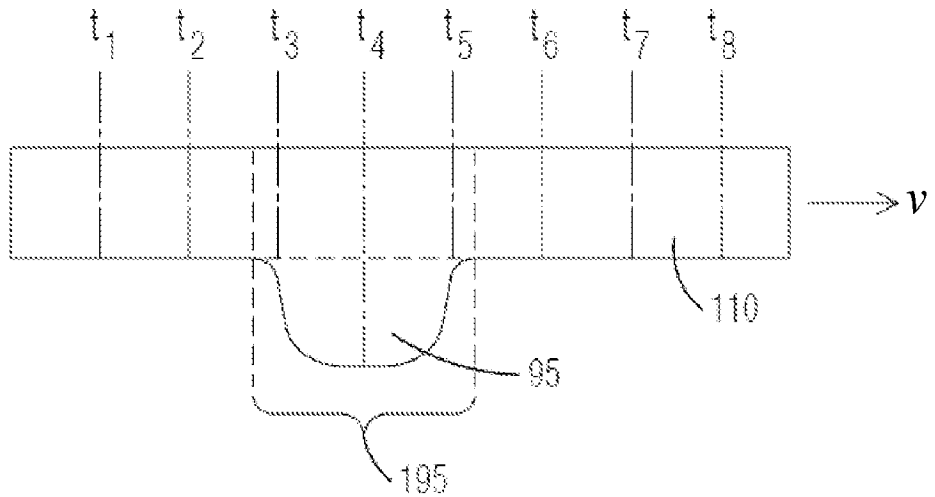


FIG. 7

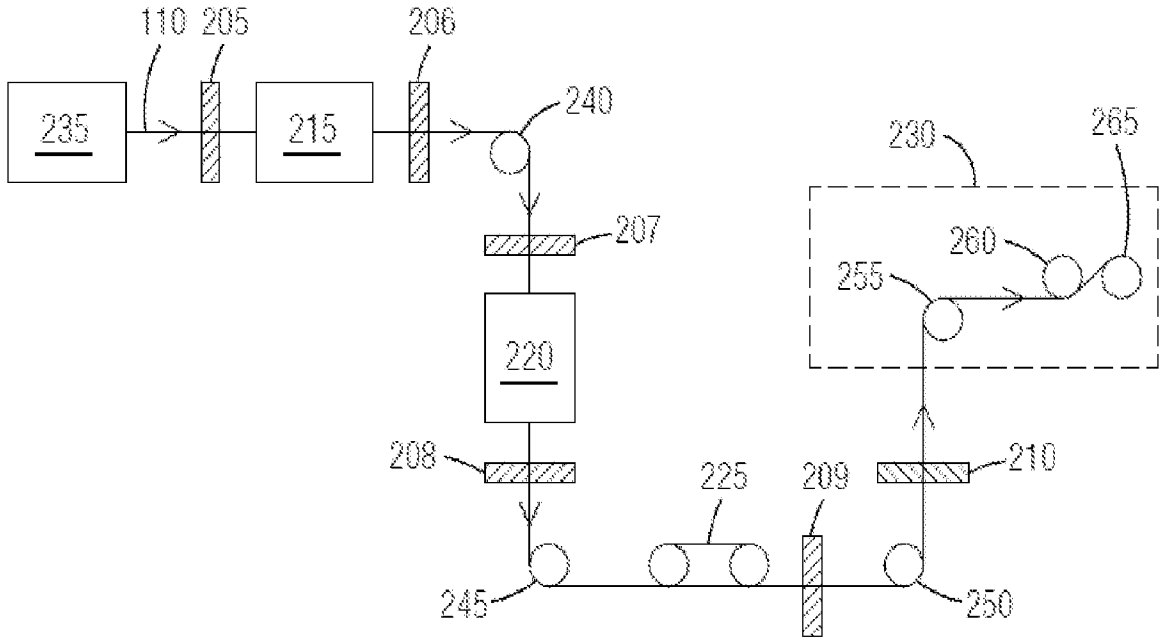
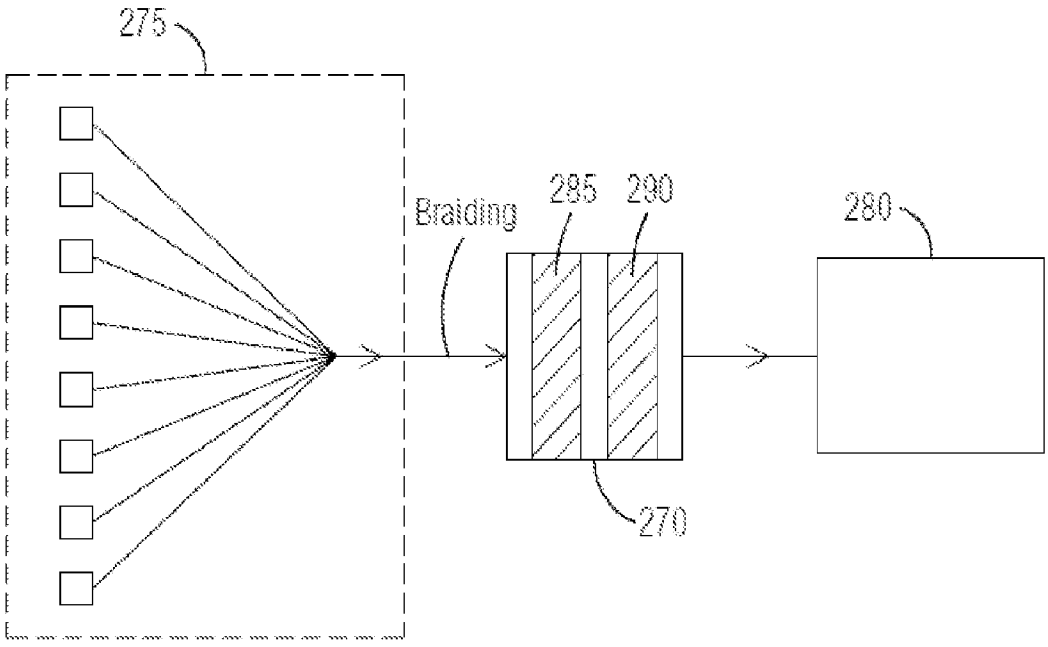

110
205
206
235
215
240
230
265
207
260
255
220
208
225
209
210
245
250

FIG. 8


275
Braiding
285
290
280
270

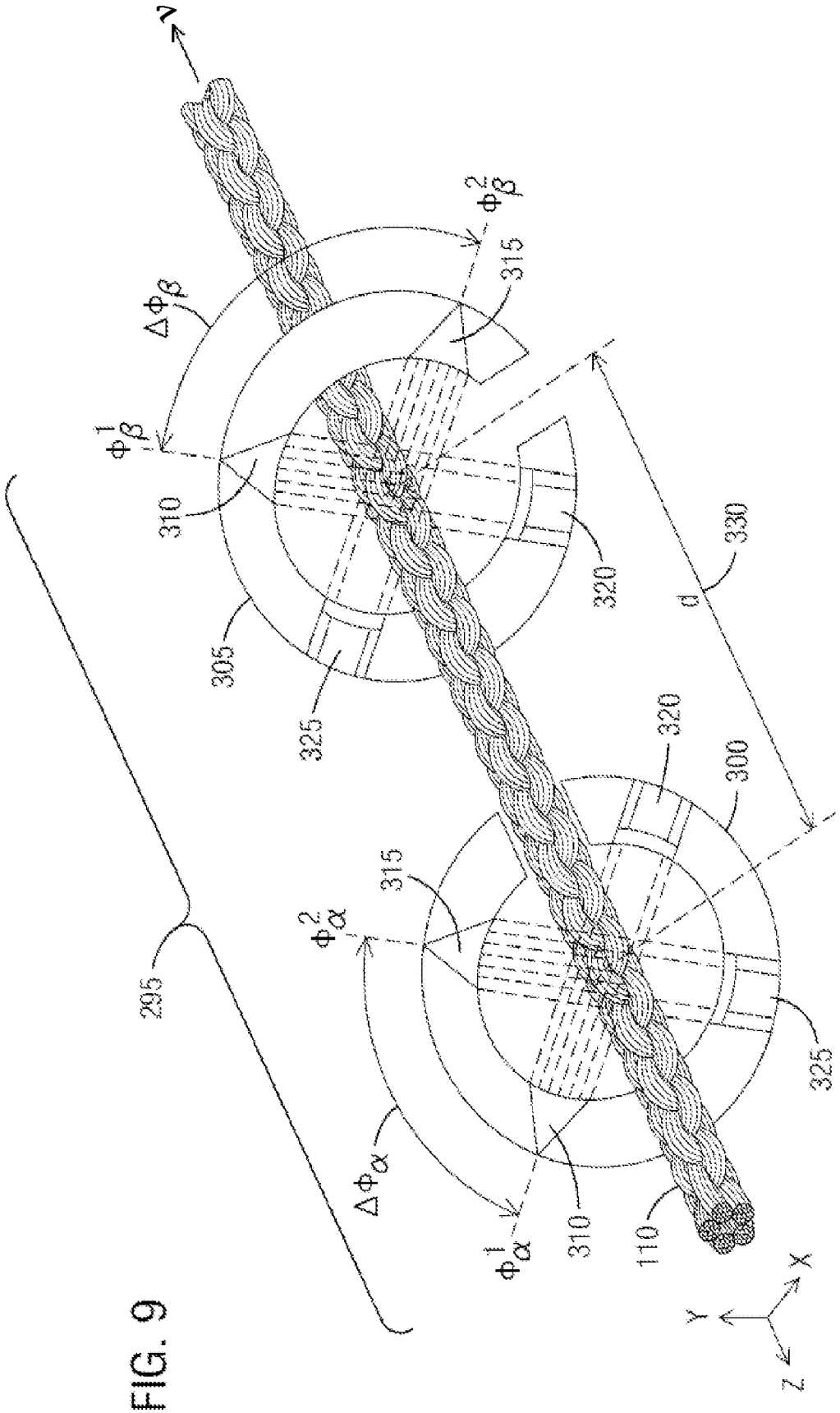

v
ϕ_β^2
315
$\Delta\phi_\beta$
ϕ_β^1
310
305
325
320
330
d
320
300
315
ϕ_α^2
295
325
$\Delta\phi_\alpha$
ϕ_α^1
310
110
X
Y
Z

FIG. 9